

Satellite Embedding–Based Mapping of Pacific Yew (*Taxus brevifolia*) Habitat Decline Across British Columbia, and an IUCN Red List Assessment of the Canadian Population

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Abstract

Pacific yew (*Taxus brevifolia*) is a slow-growing, shade-tolerant understory conifer that is both ecologically and culturally significant across its Pacific Northwest range. In British Columbia (BC), the species faces severe population decline driven primarily by industrial clear-cut logging, compounded by wildfire, stream erosion, and the legacy of paclitaxel (Taxol) bark extraction. Despite these pressures, no spatially explicit, province-scale assessment of yew habitat extent or decline has previously been undertaken.

We present a machine-learning framework using 64-band satellite spectral embeddings from the Google AlphaEarth Foundation model, classified by an XGBoost ensemble (AUC-ROC 0.996), to map Pacific yew habitat probability at 10 m resolution across 9,800 km² of BC spanning three major biogeoclimatic zones: Coastal Western Hemlock (CWH), Interior Cedar–Hemlock (ICH), and Coastal Douglas-fir (CDF). Post-classification suppression using BC Vegetation Resources Inventory stand-age records, historical fire perimeters, and digital elevation data quantifies both current remaining and historically destroyed habitat.

We estimate that 154,483 ha of yew habitat existed historically across 98 study tiles, of which 49,663 ha (32.1%) remains today — a 67.9% decline (95% CI 52–75%, tile bootstrap; conditional on the sampled tiles). Calibrating from 461 deduplicated FAIB permanent sample plot records (conditional mature-stem density 10 stems/ha, median), this habitat supports an estimated ~497,000 mature individuals, far exceeding IUCN Criterion C thresholds but carrying a large-tree size structure consistent with historical over-harvest. The CDF zone has suffered near-total loss (99.1% decline), followed by ICH (74.7%) and CWH (69.1%). This estimated loss of 104,820 ha is overwhelmingly attributable to industrial logging, and only 5.6% of mapped yew habitat falls inside provincial parks — rising to 11.0% when conservancies and national parks are included. Field-measured population size structure ($n = 120$ trees) reveals a significant deficit of large-diameter adults (DBH > 30 cm: 7 observed vs. ~32 expected under a de Liocourt stable-population model; binomial $p \approx 4 \times 10^{-9}$), independently corroborated by the FAIB inventory ($n = 461$ deduplicated stems; $p \approx 1.2 \times 10^{-43}$), consistent with selective removal of large individuals during the 1989–1993 taxol bark harvest.

Applying IUCN Red List criteria to the Canadian BC population, we find that the documented habitat decline supports a classification consistent with Endangered (EN A2c) — a significant upgrading from the current global Near Threatened (2013) listing. Because this assessment is inferred from modelled habitat decline rather than direct population census, we present it as evidence for reassessment rather than a formal listing. On the same basis, the CDF and ICH subpopulations individually reach Critically Endangered thresholds. We recommend a formal COSEWIC and IUCN reassessment and urge integration of Pacific yew habitat into forest stewardship planning across the CWH and ICH zones.

1. Introduction

Pacific yew (*Taxus brevifolia*) is a gymnosperm tree that occurs in the understorey of coniferous and deciduous forests throughout the Pacific Coast, from the southern tip of Alaska south through British Columbia into northern California. In BC, its range extends into the Interior Cedar–Hemlock (ICH) zone east of the Coast Mountains (Woods 2012; Pojar & MacKinnon 1994). It is a long-lived, dioecious conifer that occurs primarily as a scattered understorey tree in moist, old-growth forests of the Coastal Western Hemlock (CWH), ICH, and Coastal Douglas-fir (CDF) biogeoclimatic zones (Pojar et al. 1991; Meidinger & Pojar 1991).

Pacific yew plays a crucial role in coastal and riparian ecosystems, providing habitat and supporting biodiversity (Bolsinger & Jaramillo 1990). It is also culturally significant to Indigenous communities across its range, particularly in medicine, tool-making, and spiritual practices (Turner 1998, 2021; Turner & Hebda 2012). Despite its ecological and cultural importance, the species has faced multiple compounding threats throughout the twentieth century.

The species gained worldwide attention when the National Cancer Institute discovered in the 1960s that its bark contained paclitaxel, a potent mitotic inhibitor effective against ovarian, breast, and lung cancers (Wani et al. 1971). Because paclitaxel yield is extraordinarily low — approximately 1 kg per 9,080 kg of bark — and extraction required killing the tree, a "yew rush" in the early 1990s resulted in the destruction of hundreds of thousands of mature Pacific yew trees across the Pacific Northwest (Hartzell 1990). The crisis was largely averted by 1993 when semi-synthetic production from European yew (*Taxus baccata*) needles eliminated the need for wild bark harvest.

Taxus brevifolia is currently listed on the IUCN Red List as Near Threatened (Thomas 2013), with population decline attributed to historical bark harvesting and continuing industrial logging. This has resulted in a near 30% reduction in global population that is still considered to be declining. However, the primary ongoing threat to Pacific yew populations is industrial clear-cut logging. As a shade-obligate species requiring old-growth canopy conditions, Pacific yew is eliminated entirely from clearcut blocks and requires over a century to recolonise (Busing et al. 1995). It is particularly sensitive to forest fragmentation, making it highly dependent on old-growth ecosystems (Arsenault & Bradfield 1995; Reynolds 2022). Studies confirm that Pacific yews rarely re-establish in disturbed environments, and their scarcity of fruiting and low seedling survival rates further hinder recovery (Hartzell 1990; Council of the Haida Nation 2016). The species "exhibits slow recovery from major disturbances, potentially requiring centuries to return to pre-harvest abundance" (Busing et al. 1995). Pacific yew's very thin bark (approximately 0.5 cm) makes the tree highly sensitive to fire, hindering re-growth and re-establishment (Reynolds 2022; Busing et al. 1995). The increasing occurrence of high-intensity forest fires associated with climate change represents an additional negative population factor.

The remaining old-growth forests across BC are increasingly at risk from logging and climate change (Bergeron & Fenton 2012; Lindenmayer et al. 2012; Woods 2012). As the life processes of Pacific yew operate on a much longer time scale than modern forestry practices, which favour short harvest rotations and extensive clear-cutting, the yew struggles to re-establish itself once old-growth forest is cleared (Arsenault & Bradfield 1995; Council of the Haida Nation 2016).

Within this context of negative pressures, there is also a recognised lack of inventory data to understand the Pacific yew's current population status. There are acknowledged knowledge gaps and a lack of hard data on the species' current viability and distribution in BC (Reynolds 2022). This is a barrier to understanding the risks to the species and hinders informed decision-making by the provincial government and other stakeholders. An understanding of the species' population levels would also partly address First Nations concerns over the historical overharvesting of Pacific yew (Council of the Haida Nation 2016; Turner & Hebda 2012).

No comprehensive spatial assessment of Pacific yew habitat extent or decline has been attempted at the provincial scale. Existing range data consist of point observations (e.g., iNaturalist, herbarium records) supplemented by incidental mentions in forest inventory plots. This study addresses that gap using satellite-based spectral embeddings from a recently released deep learning foundation model and machine learning. The objective is to model the spatial distribution and quantify the decline of Pacific yew habitat across its primary BC range — and to apply the resulting habitat decline estimates to a formal IUCN Red List assessment of the Canadian BC population.

2. Methods

Study Rationale

While population estimates have been produced by the US Forest Service from forest inventory data in the 1990s (TODO cite), the BC Forest Analysis and Inventory Branch (FAIB) permanent sample plot (PSP) network records 461 unique Pacific yew (*Taxus brevifolia*, species code TW) stems across 122 plots province-wide. These plots are not stratified by BEC subzone in proportion to yew abundance, and the majority fall in CWH stands dominated by other conifers; the number of yew records per subzone is insufficient to separately parameterise population dynamics for CWH, ICH, and CDF. The FAIB data were therefore used here to calibrate conditional stem density (where yew is present; §4.6) rather than as a direct population model. The aim of this study was to combine spatially explicit habitat mapping with FAIB-calibrated density to produce a contemporary, province-scale population estimate using the most current remote sensing tools.

Thus, this study was based on 10 m resolution spectral features extracted from the Google AlphaEarth Foundation model satellite embeddings (Brown et al. 2025) combined with iNaturalist crowd-sourced occurrence data, which was used to build an XGBoost (TODO cite xgboost) classifier to give a probability of finding a yew tree in a given 10 m × 10 m pixel. This probability map was then adjusted based on known logging, fire, and riparian erosion criteria to give a final yew probability map, from which total habitat area is estimated as the probability-weighted sum of pixel areas (probability mass; §2.5.4). Multiplying this habitat area by the FAIB-calibrated conditional mature-stem density provides a population estimate of the order of ~497,000 mature individuals (§4.6).

Computational constraints limited the scope of this survey to 98 tiles selected to represent the diversity of BC yew habitat, rather than exhaustively covering the entire provincial range. Tiles, rather than individual pixels, were used as the unit of analysis to display the geographic predictions of the model, better enabling assessment of whether the model agreed with known habitat associations of the yew, such as higher densities near rivers and riparian zones and low densities in disturbed areas.

2.1 Study Area

The 98 study tiles (each ~10 × 10 km, ~100 km²) were distributed across British Columbia (Figure 3), covering a total of approximately 9,800 km². Of these, 85 tiles were located in coastal BC within the CWH and CDF zone, and 14 tiles were placed within the Interior Cedar–Hemlock (ICH) zone in the BC interior. Although the center of a tile would be in either the ICH, CDF, or CWH zone tile areas would sometimes sample adjacent zones. A full list of zones and areas sampled can be found in Table 1.

2.1.1 Coastal Tile Selection

The 85 coastal tile centre pixels were selected by random stratified sampling within the CWH and CDF biogeoclimatic zones. The 14 ICH tiles were generated by random stratified sampling within the ICH

biogeoclimatic zone. Boundary polygons for the zones were from the BC BEC map, v12. Sampling used a fixed random seed (42) projected in BC Albers and converted to WGS84.

Table 1.

Across all 98 tiles, the analysis intersects 12 major BEC zones and 69 subzones:

Zone	Full Name	Tiles with Coverage	Total Area Analysed (ha)
CWH	Coastal Western Hemlock	85	885,783
ICH	Interior Cedar–Hemlock	14	142,778
CDF	Coastal Douglas-fir	5	36,055
ESSF	Engelmann Spruce–Subalpine Fir	25	70,798
MH	Mountain Hemlock	20	99,030
IDF	Interior Douglas-fir	5	8,595
CMA	Coastal Mountain-heather Alpine	10	147,304
MS	Montane Spruce	4	9,189
IMA	Interior Mountain-heather Alpine	3	9,789
BAF	Boreal Altai Fescue Alpine	1	5,901
SBS	Sub-Boreal Spruce	1	793

2.2 Satellite Spectral Embeddings

2.2.1 Google AlphaEarth Foundation Model

Per-pixel spectral features were extracted from the Google AlphaEarth Foundation model satellite embeddings ([GOOGLE/SATELLITE_EMBEDDING/V1/ANNUAL](#)), accessed through Google Earth Engine (GEE). This dataset provides 64-dimensional embedding vectors (bands A00–A63) at 10 m spatial resolution, derived from annual Sentinel-2 composites using a deep neural network trained on global satellite imagery. We used the 2024 annual composite for all tiles.

The embeddings capture complex spectral-spatial patterns related to vegetation structure, moisture regime, canopy density, and land cover type. Each embedding dimension is approximately unit-normalised, consistent with the AlphaEarth Foundation model design.

2.3 Training Data

2.3.1 Positive Samples

Yew presence records comprised 1,043 research-grade *Taxus brevifolia* observations from iNaturalist (BC and Washington/Oregon), partitioned into 834 training and 209 validation records. iNaturalist records were retained only where the coordinate showed no sign of development or logging; although some yew trees persist after logging, these represent a small minority of logged area in the province and their mortality in plantations is high, and urban locations no longer reflect habitat in which yew grows naturally, both classes were therefore excluded from training to avoid confusing the model. Centre-pixel 64-dimensional embeddings were extracted at each observation location, yielding per-observation feature vectors of length 64 representing the spectral embedding at the pixel containing the GPS coordinate. An additional 64 field-verified positive observations from southern Vancouver Island were included in the positive training set, giving 1,107 combined positive records.

2.3.2 Negative Samples

Negative (yew-absent) training data comprised 11,452 samples drawn from: - BC Forest Analysis and Inventory Branch (FAIB) forest plots in non-yew species assemblages - Alpine and subalpine locations above the elevational range of yew - CWH-zone locations with confirmed non-yew canopy dominance (western hemlock, Sitka spruce stands)

The combined training set therefore comprised 12,559 unique records (1,107 positives and 11,452 negatives), with the GEE-derived negatives duplicated 2× at training time for an effective negative weight of 20,446.

2.4 Classifier Architecture

2.4.1 Production XGBoost Model

The production classifier is an XGBoost gradient-boosted tree ensemble trained on the 64 raw embedding dimensions with the following hyperparameters:

Parameter	Value
Objective	binary:logistic
Eval metric	AUC
Max depth	6
Learning rate	0.05
Subsample	0.8
Column sample by tree	0.8
Number of boosting rounds	500 (early stopping at 50)

The model was trained with spatial cross-validation using K-means clustering (10 clusters) for train/validation/test splits (70/10/20%) to mitigate spatial autocorrelation.

2.4.2 Comparison Models

Six additional classifiers were evaluated on identical features and splits (Figure 4):

Model	AUC-ROC	Accuracy	F1 Score
XGBoost (production)	0.996	0.989	0.947
MLP + StandardScaler	0.9961	0.986	0.977
MLP raw embeddings	0.9962	0.976	0.960
Random Forest	0.9896	0.984	0.974
kNN (k=3)	0.9909	0.911	0.833
Logistic Regression	0.9165	0.813	0.562

The production XGBoost AUC-ROC of 0.996 is statistically indistinguishable from the best-performing neural models. XGBoost was selected as the production model for its comparable performance, interpretability, and computational efficiency for per-pixel inference across ~100 million pixels. Although the MLP achieved comparable held-out performance, its propensity for overfitting, evident as noise in the spatial distribution of its classifications, discouraged its use on generalizability grounds.

To test whether the reported AUC reflects discrimination within ecologically similar stands rather than the trivially easy separation of coastal old-growth from alpine or boreal forest, we evaluated the production model

against a same-BEC-zone negative set: 2,811 non-yew locations drawn exclusively from CWH, ICH, and CDF zones (the same zones where positives occur), compared with 3,383 "easy" negatives from spectrally dissimilar zones (SBS, IDF, BWBS, SBPS, PP). AUC declined only marginally between the easy (0.9998) and hard (0.9992) negative sets, but the false positive rate within coastal BEC zones was 17× higher (1.0% vs 0.06%), indicating the model does not purely rely on zone identity. These comparisons use the same training distribution (in-sample) and therefore overestimate absolute performance; their value is directional. A fully held-out hard-negative evaluation would require new GEE embeddings beyond the current dataset scope (`scripts/analysis/hard_negative_eval.py`; results in `results/analysis/hard_negative_eval.json`).

2.4.3 Alternative Feature Sets

A separate XGBoost model trained on 35 engineered features from the BC Vegetation Resources Inventory achieved AUC 0.82 with only 26% recall at operational thresholds, and a multi-modal model combining spectral embeddings with inventory features yielded only 5% recall due to scale mismatch. These results confirmed that satellite spectral embeddings combined with iNaturalist data, alone provided sufficient classification performance.

2.4.4 Independent map validation

The final suppressed-probability habitat map was validated against two datasets that were independent of both classifier training and tile selection: (1) the held-out iNaturalist validation split (208 BC positives; 6 matched study tiles) and (2) FAIB PSP sites (2,911 sites; 167 matched tiles, 4 with *T. brevifolia*). All 6 iNat validation-set positives within study tiles, and all 4 yew-present FAIB PSP sites within tiles, occurred in VRI logging categories 1–4 (water or logged stands), where the suppression pipeline correctly assigns near-zero probability. No independent yew-present occurrence records within study tiles coincided with undisturbed old-growth (VRI cat 7), preventing a formal sensitivity estimate for the unsuppressed map. The Boyce index — the standard presence-only validation metric — could not be computed reliably with fewer than 10 matched presence points (`scripts/analysis/map_validation_boyce.py`; full output in `results/analysis/map_validation_boyce.json`). This limitation reflects the map's purpose: it depicts old-growth yew *habitat*, not the location of every surviving individual; yew in logged stands (leave-trees) exist but do not constitute habitat in the recruitment sense the map models.

2.5 Post-Classification Suppression

Raw XGBoost probabilities represent spectral habitat suitability but do not account for disturbance history. We applied a sequential suppression pipeline to convert raw probabilities into ecologically realistic current habitat estimates.

2.5.1 Logging Suppression

Stand-age information was extracted from the BC Vegetation Resources Inventory (VRI 2024, 6.87 million polygons). Each 10 m pixel was assigned to one of seven land-use categories:

Category	Description	Suppression Factor
1	Water / non-forest	×0.00
2	Logged <20 yr	×0.00
3	Logged 20–40 yr	×0.00
4	Logged 40–80 yr	×0.20
5	Forest 80–150 yr	×0.50

Category	Description	Suppression Factor
6	Alpine / barren	×0.00
7	Old-growth (>150 yr)	×1.00

Categories 2 and 3 (logged < 40 yr) and category 6 (alpine) are suppressed to zero, reflecting the absence of yew in recently cleared or non-forested stands. Category 4 (logged 40–80 yr) receives a ×0.20 factor, acknowledging that occasional yew leave-trees or root-sprouting survivors may persist in older second-growth but that stand structure is unsuitable for recruitment. Category 5 (forest 80–150 yr, maturing second-growth) receives a ×0.50 factor: stands of this age may carry residual yew from the pre-harvest cohort, but canopy closure and altered light conditions suppress recruitment below replacement. Only category 7 (old-growth >150 yr) is treated as fully suitable (×1.00). For habitat *loss* estimation the historical baseline uses the raw model probability uniformly across all forested land — old-growth and logged alike — weighted by the BEC-subzone yew prevalence rate (§2.5.4), so that the partial suppression factors apply only to the *current* estimate. This assumption likely inflates the historical baseline: not all forested land that was subsequently logged would have supported yew under old-growth conditions, so the 154,483 ha figure is best interpreted as an upper bound on original suitable habitat rather than a precise census of former range.

2.5.2 Fire Suppression

Historical fire perimeters (1900–2024) from the BC historical fire polygon dataset were filtered to fires ≥100 ha. Rather than an arbitrary linear time-since-fire ramp, we applied a fire modifier with an explicit burn severity and a demographically grounded recovery rate. Within a fire perimeter we assume 75% of the yew habitat is killed (Pacific yew has ~0.5 cm bark and essentially no fire tolerance, so burned area implies near-total local mortality) and 25% survives in unburned refugia and skips. The killed fraction then recovers along the Busing & Spies (1995) stage-projection matrix for old-growth yew ($\lambda = 1.02$): starting from a reseeded seedling stand, the old-growth large-tree cohort (stages 6–7, >15 cm DBH — the trees that define old-forest yew habitat) rebuilds only as seedlings grow through the intervening size classes. The surviving fraction of pre-fire habitat at $t = 2024 - \text{fire_year}$ years post-fire is

$$\text{fire_modifier}(t) = (1 - 0.75) + 0.75 \cdot R(t)$$

where $R(t)$ is the Busing-matrix recovery fraction of the large-tree cohort ($R(0) = 0$, rising to ~0.9 only by $t \approx 105$ yr). This reproduces Busing & Spies' conclusion that disturbed stands need centuries to recover the population size and structure characteristic of old-forest stands. For overlapping polygons, the most recent fire year takes precedence.

2.5.3 Elevation Suppression

A Copernicus GLO-30 Digital Elevation Model was used to suppress yew predictions at very low coastal elevations:

$$\text{elev_factor} = \text{clip}((\text{elevation}) / (30), 0, 1)$$

This linear ramp from 0 at sea level to 1 at 30 m removed 18,434 ha of false-positive predictions from intertidal and low-coastal zones. The 30 m threshold is applied as a conservative heuristic rather than a hard ecological boundary: Pacific yew is not salt-tolerant and does not establish in the intertidal, foreshore, and saline-influenced floodplain settings that dominate below this elevation (Busing et al. 1995), where the embedding classifier nonetheless returns occasional high probabilities driven by spectrally similar moist, closed-canopy vegetation. The linear ramp avoids a sharp cut-off and down-weights, rather than eliminates, predictions in the transitional 0–30 m band.

2.5.4 Habitat Loss Estimation

For each BEC subzone, yew prevalence rate (r) was computed as the mean raw XGBoost probability across all old-growth pixels. This yew prevalence rate was used as an estimator of the original density of yews in the subzone. Estimated original habitat was $r \times (\text{old-growth} + \text{logged pixels}) \times 0.01 \text{ ha/pixel}$. The factor 0.01 ha/pixel converts pixel counts to area: at the 10 m native resolution of the embeddings, each pixel covers $10 \text{ m} \times 10 \text{ m} = 100 \text{ m}^2 = 0.01 \text{ ha}$. Here and throughout, "probability mass" denotes the sum of per-pixel yew probabilities (each in $[0, 1]$) multiplied by this per-pixel area — equivalently, the area-weighted expected habitat extent. This approach uses continuous probability mass rather than binary thresholding (counting only pixels above a fixed probability cut-off), providing a more robust cumulative habitat-area estimate that is insensitive to the choice of threshold.

2.6 Secondary Threat Assessment

In addition to the quantitative logging and fire analysis, we reviewed the scientific literature to characterise secondary threats not directly modellable from remote sensing.

2.6.1 Stream Erosion and Riparian Habitat Loss

Pacific yew preferentially occupies moist riparian zones (Busing et al. 1995). Logging-driven hydrological changes increase peak flows by 20–50% (Hartman & Scrivener 1990), causing channel widening proportional to $W \propto Q^{0.5}$ (Leopold & Maddock 1953). A water buffer sensitivity analysis was run across all 41 tiles with available grid data: applying binary morphological dilation (3 pixels = 30 m) to all water category pixels, then summing yew probability mass in the buffered old-growth pixels. This yielded 1,717 ha of yew probability mass at risk from riparian erosion (5.9% of the 29,028 ha of remaining yew probability mass in those 41 tiles — a subset of the full 49,663 ha, restricted to the tiles for which the VRI water-category grid was available).

2.6.2 Sea-Level Rise and Saltwater Intrusion

Pacific yew being sensitive to salt; saline groundwater intrusion kills root systems. Habitat below 1.0–1.5 m elevation in 50-year projections faces 100% loss, with an additional 15–20 m inland buffer for saltwater intrusion effects. Our elevation suppression (§2.5.3) captures current low-elevation effects but does not project future sea-level scenarios.

2.6.3 Yew Big Bud Mite

The Yew Big Bud Mite (*Cecidophyopsis psilaspis*) causes bud galls on *Taxus* species, with terminal bud mortality averaging over 20% in infested coastal BC populations, reducing growth rates by ~20% and seed (aril) production by ~25% (TODO find citation). No *T. brevifolia*-specific dose-response curve was found in the academic literature or provincial forest health reports; the 20%/25% multipliers used in the population viability analysis (§3.4) are therefore illustrative sensitivity bounds rather than empirically fitted values and should be interpreted accordingly.

2.6.4 Ungulate Browsing

Wild ungulates preferentially browse Pacific yew foliage in winter. Browsing pressure causes 60–80% seedling/sapling mortality in areas with high ungulate density ($>10 \text{ deer/km}^2$), creating a "browsing ceiling" that prevents recruitment to the established understorey stage (Busing et al. 1995; Council of the Haida Nation 2016).

2.6.5 Wildfire Frequency Increase

Climate change is projected to reduce fire return intervals in drier maritime subzones to 80–120 years, potentially preventing populations from reaching reproductive maturity (~80–100 years) between successive fires.

2.6.6 Historical Taxol Bark Harvest

Between 1989 and 1993, at peak harvest (1991), an estimated 360,000 mature yew trees per year were felled across the Pacific Northwest for paclitaxel extraction. Treating a single cancer patient required the bark of six 100+-year-old trees. This harvest was not spatially documented in GIS and cannot be directly modelled, but its legacy is embedded in the depleted large-tree cohort documented in §3.10.

2.7 Field Sampling of Population Size Structure

To ground-truth population age structure independently of remote sensing, diameter at breast height $n = 120$ Pacific yew individuals along East Muir Creek in January 2026. All yew stems ≥ 1 cm CBH within 30 m of the stream bank were measured along a 3 km portion of the river.

Size-class structure was compared against the de Liocourt reverse-J model, which describes the expected stem-frequency distribution in a balanced, self-sustaining uneven-aged stand as a constant ratio q of stems between successive (ascending) diameter classes. We use this model as a descriptive reference for a stable understory population rather than as a mechanistic demographic model: because Pacific yew is a slow, plastic, suppression-tolerant grower, diameter is an imperfect proxy for age, and we therefore frame the comparison in terms of size structure, not age structure. For Pacific yew in intact old-growth CWH and ICH stands, Bolsinger & Jaramillo (1990) and Graham (1994) document $q \approx 1.4$ – 1.6 per 10-cm class; we adopt $q = 1.5$ as the central reference and report all comparisons across the full 1.4–1.6 range so that conclusions do not depend on a single assumed value. Expected stem counts per 10-cm class were obtained by scaling the geometric series q^{-i} to the observed sample size.

Three complementary tests were applied. First, because the *a priori* prediction of selective large-tree removal concerns the large-diameter tail, we tested the count of stems > 30 cm DBH against the proportion expected under each reference q using an exact binomial test (one-sided, deficit). Second, a Pearson χ^2 goodness-of-fit test was applied to the binned counts (the three classes below 30 cm plus a pooled ≥ 30 cm class, so that all expected counts exceeded five). Third, to summarise the realised decline rate we fitted a single reverse-J slope to the whole observed distribution by ordinary least squares on the log-counts of all occupied 10-cm classes, yielding an empirical population q with a 95% confidence interval from a non-parametric bootstrap (2,000 resamples of the $n = 120$ stems); this whole-population q — unlike a fit restricted to the sparse adult tail — is well constrained and its direction relative to $q = 1.5$ is interpretable. Two caveats bound all of these tests: (i) understory stems < 10 cm DBH are readily overlooked, so the smallest class may be under-counted; and (ii) multi-stemmed (layering) individuals were recorded as separate stems (§2.7), which can inflate the small-stem classes relative to the genet-based assumptions of the de Liocourt model.

2.8 Interactive Web Map

Results are presented via an interactive web map hosted on GitHub Pages (jerichooconnell.github.io/yew_project). The map displays yew probability overlays for all 98 tiles, VRI-derived logging age class rasters, 5,700 historical fire polygons ≥ 100 ha (1900–2024) colour-coded by age, 1,201 protected areas (provincial parks, conservancies, national parks, ecological reserves, and protected areas, from the BC Data Catalogue), and a crowd-sourced field observation reporting interface.

3. Results

3.1 How much yew habitat remains, and where?

Across all 98 study tiles (~9,800 km², 69 BEC subzones), we estimate that 154,483 ha of yew habitat existed historically under pre-logging old-growth conditions, of which 49,663 ha (32.1%) remains today (Table 1, Figure 1). This represents an overall decline of 67.9% (95% CI 52–75% from a 10,000-replicate bootstrap resampling the 35 tile-matched controls; `scripts/analysis/decline_bootstrap_ci.py`), with per-zone severity ranked in Figure 1. The lower bound of this interval still exceeds the 50% threshold for IUCN Endangered status (§4.6). The interval reflects between-tile variance only; it is conditional on the purposive tile sample and on the modelled old-growth baseline, and does not propagate uncertainty in the baseline itself. Unless otherwise stated, all headline habitat areas are *continuous probability-mass* estimates (the sum of per-pixel yew probability × 0.01 ha; §2.5.4) rather than counts of pixels above a fixed threshold; threshold-based figures used in specific secondary analyses (37,858 ha at $P \geq 0.5$ and 29,028 ha in the 41-tile erosion subset) are defined where they appear.

The remaining 49,663 ha is not evenly distributed. The Coastal Western Hemlock zone holds 72% of it (34,386 ha), concentrated in remote hypermaritime terrain where logging pressure has been lower. The Interior Cedar–Hemlock zone holds a further 13% (6,385 ha), mostly in isolated valley-bottom fragments. At the opposite extreme, the Coastal Douglas-fir zone — which historically supported some of the densest yew populations on Vancouver Island and the Gulf Islands — retains only 36 ha. The geographic pattern of habitat, logging pressure, old-growth refugia, and protection across the study tiles is summarised in Figure 8: logging concentrates in the southern coast and interior, the largest old-growth refugia persist on the central and north coast, and protected areas are sparse and poorly aligned with the richest remaining habitat. The underlying per-tile model output is shown directly in Figures 6 and S4, which place the predicted-probability rasters at their true locations across Vancouver Island and the Central and North Coast respectively.

The model pipeline that converts raw per-pixel spectral probabilities to this final estimate applies three sequential corrections (Figure 2): logging suppression (removing predictions from stands <150 yr old, –146,506 ha; the correspondence between predicted probability and logging status is shown for a representative tile in Figure 5), fire suppression (a 75%-burn modifier with Busing-matrix recovery on historical burn perimeters, ~–700 ha), and elevation correction (removing false positives below 30 m elevation, –18,434 ha). The elevation correction is a model accuracy adjustment, not a quantity of ecological loss, and is excluded from all habitat-decline accounting. Mean yew probability in old-growth pixels is highest in warm, moist subzones at low to mid elevation (ICHxw: 0.489; CWHvh1: 0.411; ICHdw1: 0.410; CDFmm: 0.381), reflecting the species' preference for frost-free, moisture-retentive sites.

Table 1. Summary statistics by major BEC zone (zones with >10 ha estimated original yew habitat).

Zone	Est. Original (ha)	Current Remaining (ha)	Decline (%)	Logged Area (ha)	Fire Loss (ha)
CWH	111,407	34,386	69.1	312,128	460
ICH	25,257	6,385	74.7	97,071	152
CDF	3,889	36	99.1	9,986	0
ESSF	3,957	1,742	56.0	29,392	64
MH	7,078	4,161	41.2	21,709	7
IDF	1,697	463	72.7	5,768	0
MS	552	114	79.4	6,511	6

Zone	Est. Original (ha)	Current Remaining (ha)	Decline (%)	Logged Area (ha)	Fire Loss (ha)
CMA	609	225	63.0	10,474	1
Total	154,483	49,663	67.9	493,039†	692

†The "Logged Area" column reports total logged *forest* area within each zone (all non-old-growth forested land); the column sums to 493,039 ha but this total is distinct from the area of logged *yew habitat* (~104,820 ha = original – remaining). Zone totals reflect overall forest disturbance, not yew-specific loss.

3.2 What drove the decline?

3.2.1 Logging as the primary driver

Industrial clear-cut logging accounts for an estimated 104,820 ha of destroyed yew habitat — approximately 99.4% of all ecological loss documented in this study. This dominance is reflected in the suppression pipeline: the logging filter removes 146,506 ha (68.7%) of raw model predictions, far exceeding the contributions of fire (~700 ha) and all secondary threats combined. The three focal zones exhibit qualitatively different logging-driven decline pathways:

Coastal Western Hemlock (CWH): The CWH zone was logged extensively from the 1940s through the 1980s, with the 40–80 yr stand-age class now dominant (125,262 ha). Despite retaining 37.3% old-growth by area, 35.2% of the landscape is logged, and decline rates exceed 83% in the drier maritime subzones most intensively harvested (CWHxm2: 89.5%; CWHmm2: 90.7%; CWHdm: 83.9%). The CWHvh2 subzone retains the most yew habitat in absolute terms (10,355 ha) because its remote hypermaritime terrain (121,647 ha old-growth) has been less accessible to industrial forestry (Figure S1).

Interior Cedar–Hemlock (ICH): The ICH zone shows the second-highest overall decline (74.7%), exceeding the CWH despite a smaller absolute area, because 68.0% of the ICH landscape has been logged versus 35.2% of the CWH. Yew habitat in the ICH concentrates in accessible valley bottoms — precisely the terrain most intensively harvested. The ICHxw subzone has the highest raw yew probability in old-growth of any sampled subzone (0.489), yet has lost 83.8% of its estimated original habitat to logging. Recent logging (<20 yr stands: 22,579 ha, 23% of total logged ICH area) indicates that harvest pressure remains active (Figure S2).

Coastal Douglas-fir (CDF): The CDF zone represents a qualitatively different, more extreme trajectory: near-complete regional extirpation. Of 36,055 ha analysed, only 214 ha of old-growth remains (0.6%), and modelled current habitat is just 36 ha — a 99.1% decline. The CDFmm subzone historically had the highest yew prevalence of any sampled subzone (0.381), confirming that the CDF was prime yew habitat. Its near-total loss reflects not only commercial logging but decades of urban expansion and agricultural conversion across southern Vancouver Island and the Gulf Islands.

3.2.2 Fire as a secondary driver

Across all tiles, 5,700 historical fire perimeters (1900–2024) covering approximately 96,543 ha intersected the study area, yielding an estimated ~700 ha of yew habitat suppression in remaining old-growth stands — small relative to logging but non-negligible in drier interior subzones (ICH: ~150 ha; CWH: ~465 ha; per-zone values in Table 1 are from the earlier linear model and sum to 692 ha). Fire suppression is estimated with a 75%-burn-severity modifier whose post-fire recovery follows the Busing & Spies (1995) stage-projection matrix (§2.5.2): a burned stand stays largely suppressed for decades and rebuilds old-growth structure only over ~century timescales, so older fires retain only partially recovered yew habitat while 25% survives in unburned refugia within each perimeter. This physically-grounded model gives essentially the same total as the earlier linear modifier (692 ha), confirming the fire estimate is robust to the suppression-model choice. The 2020s burned

area (8,146 ha) exceeds the 2010s (4,859 ha), consistent with an accelerating fire season driven by climate change, suggesting that fire's contribution to future habitat loss will grow.

Because fire is applied to the logging-suppressed grid in the pipeline, the ~700 ha figure captures only fire's effect on the old-growth that survived logging; fire's effect on habitat that was *also* logged is invisible to it. To isolate fire's contribution independent of logging, we applied the same 75%-burn Busing-recovery modifier to the raw (pre-logging) probability grid across the full original-habitat footprint — old-growth plus logged forest — for the 41 tiles with cached pixel data, then scaled to all 98 tiles by the ratio of historical habitat mass (`scripts/analysis/fire_independent_loss.py`). This yields an estimated ~3,730 ha of yew habitat that historical fire perimeters would have suppressed absent logging — approximately 2.4% of the original 154,483 ha, and roughly five times the pipeline figure. The large majority of this loss, ~3,330 ha (89%), falls in stands that were subsequently logged ("dual-threatened" habitat affected by both fire and logging), with only ~400 ha in old-growth that escaped logging. Fire's independent footprint is therefore an order of magnitude smaller than logging's (104,820 ha) — the dominance of logging in the combined loss (>97% under either accounting) is unchanged — but it is concentrated in the drier, more fire-prone coastal and interior tiles (e.g. Strathcona Highlands, Squamish Highlands, Stave Lake), where it compounds logging pressure and raises the recovery barrier on a subset of the logged landscape.

3.2.3 Cumulative secondary threats

Beyond logging and fire, a suite of secondary threats compounds the ecological impact (Table 2). Of the quantified secondary threats, stream erosion is the most spatially significant: applying a 30 m buffer (3-pixel morphological dilation) to all VRI water-category pixels across 41 tiles with available grid data and summing yew probability mass in the buffered old-growth zone yields 1,717 ha at risk from riparian erosion — 5.9% of the 29,028 ha remaining yew probability mass in those tiles. This estimate draws on the Carnation Creek study showing 20–50% peak flow increases post-logging (Hartman & Scrivener 1990) and hydraulic geometry relationships predicting ~14% channel widening for a 30% flow increase ($W \propto Q^{0.5}$; Leopold & Maddock 1953).

Table 2. Estimated impacts of threats to Pacific yew in BC.

Threat	Estimated Impact	Confidence	Type
Clear-cut logging	104,820 ha total habitat loss	High (modelled)	Direct habitat destruction
Wildfire (historical)	~700 ha suppressed	High (modelled)	Direct mortality
Stream erosion buffer (30 m)	~1,717 ha (5.9% of remaining in sampled tiles)	Moderate	Riparian habitat loss
Sea-level rise (future)	~240 ha (<0.5% of remaining)	Low (projected)	Coastal inundation
Yew big bud mite	20–25% bud mortality	Moderate (literature)	Growth/fecundity reduction
Ungulate browsing	60–80% seedling mortality	Moderate (literature)	Regeneration failure
Historical Taxol harvest	Unknown (hundreds of thousands of trees)	Low (no spatial data)	Historical direct mortality

3.3 Are remaining populations viable? Field evidence from population size structure

Independent of the remote sensing analysis, field-measured population size structure provides a second line of evidence on species status. Measurement of $n = 120$ Pacific yew individuals yielded CBH values of 1–218 cm, corresponding to converted DBH values of 0.3–69.4 cm (mean DBH = 15.0 cm, median = 12.7 cm; Figure 9).

Size-class structure differed markedly from the de Liocourt reference distribution ($q = 1.5$), with a pronounced deficit of large-diameter stems and a surplus of mid-sized stems (Table 3).

Table 3. Observed versus expected (de Liocourt $q = 1.5$) stem counts by 10-cm DBH class.

DBH Class (cm)	Observed	Expected ($q = 1.5$)	Surplus / Deficit
0–10	36	41.6	–5.6
10–20	56	27.7	+28.3 surplus
20–30	21	18.5	+2.5 surplus
30–40	2	12.3	–10.3
40–50	4	8.2	–4.2
50–60	0	5.5	–5.5
60–70	1	3.7	–2.7
70–80	0	2.4	–2.4

†The large-tree deficit is insensitive to the assumed q : the expected count > 30 cm DBH is 38.2 ($q = 1.4$), 32.1 ($q = 1.5$), and 27.1 ($q = 1.6$), against 7 observed in every case.

Only 7 stems exceed 30 cm DBH, against 32.1 expected under $q = 1.5$ (and 27–38 across $q = 1.4$ – 1.6). An exact binomial test rejects the stable-population expectation for this tail (7 of 120 vs. expected proportion 0.27; $p \approx 4 \times 10^{-9}$ at $q = 1.5$, and $p < 10^{-6}$ across $q = 1.4$ – 1.6), as does a χ^2 goodness-of-fit test on binned counts ($\chi^2 = 49.5$, $df = 3$, $p \approx 1 \times 10^{-10}$). Fitting a single reverse-J slope to the whole observed distribution gives an empirical population $q \approx 2.0$ (95% bootstrap CI 1.6–2.7), significantly steeper than the literature-stable $q \approx 1.5$ — the realised distribution declines faster toward the large-diameter classes than a self-sustaining population, consistent with depletion of large adults. The observed distribution is not strictly monotonic (it peaks in the 10–20 cm class); this modal bulge most plausibly reflects under-detection of the smallest understorey stems and/or the recording of layering ramets as separate stems (§2.7), and does not affect the large-tree result.

The combination of abundant mid-sized stems and depleted large adults is consistent with selective harvest of large, bark-rich individuals for paclitaxel extraction during the 1989–1993 taxol rush: a recruitment bottleneck would instead deplete the smallest classes, which is not observed. We caution that this size signature is not uniquely diagnostic of harvest — an even-aged cohort that has grown into the 10–20 cm class would produce a broadly similar structure — and that the inference depends on the field sample being a representative stand census (§2.7). With that caveat, the depletion is ecologically significant because Pacific yew reaches bark-harvest size (~ 30 cm DBH) only after 150–250 years of growth (Graham 1994), making any lost large-tree cohort non-renewable on forestry time scales.

FAIB PSP comparison. The field-sample finding is independently corroborated by the provincial FAIB permanent sample plot (PSP) inventory. Among 461 *T. brevifolia* stems recorded across 122 BC plots, the large-tree deficit is still more pronounced: only 10 stems exceed 30 cm DBH against 115.4 expected under a stable $q = 1.5$ distribution (binomial $p \approx 1.2 \times 10^{-43}$; $\chi^2 = 537$, $df = 3$, $p \approx 5 \times 10^{-116}$). The whole-population fitted $q = 2.73$ (95% CI 2.13–3.70), steeper than the field-sample value of $q = 2.00$ but in the same direction and with a substantially stronger statistical signal owing to the larger sample (Figure 10). The FAIB plots are drawn from operational forest inventory, not targeted yew surveys, making their large-tree deficit unlikely to reflect collection bias toward small trees. Together, the two independent datasets agree that the province-wide large-tree cohort is severely depleted relative to stable-population expectations.

The two datasets differ in their small-diameter composition: the FAIB sample is heavily weighted toward the 0–10 cm class (median DBH 5.5 cm), while the field sample peaks in the 10–20 cm class (median 12.7 cm), suggesting proportionally fewer seedlings and saplings in the field sample sites. Two mechanisms could explain this discrepancy. First, wild ungulates (black-tailed deer, Roosevelt elk, and moose) preferentially browse Pacific yew foliage and suppress seedling and sapling recruitment in areas of high ungulate density; if the field sample sites experience greater ungulate pressure than the average FAIB plot, small stems would be systematically removed from the field-site understorey (Busing et al. 1995; Council of the Haida Nation 2016). Second, the eriophyid mite *Cecidophyopsis psilaspis* suppresses new bud growth in coastal yew populations (Reynolds 2022); repeated mite damage can prevent small individuals from growing into the 10–20 cm class, effectively suppressing the 0–10 cm cohort over time. The FAIB plots, drawn from operational forest inventory, are more broadly distributed across the province and may include sites where these regeneration stressors are less prevalent. Regardless of mechanism, the modal shift does not alter the core finding: both datasets show the same large-tree depletion relative to stable-population expectations.

To translate the observed size structure into a viability statement, we seeded the Busing & Spies (1995) stage-projection matrix with the observed (FAIB + field) stage structure and projected it forward under contrasting scenarios, including a stochastic quasi-extinction analysis (Figure S5; [scripts/analysis/yew_pva.py](#)). Vital rates in the Busing matrix are calibrated to reproduce $\lambda = 1.02$ and $q = 1.5$ for undisturbed old-growth. We independently verified stage-specific annual survival using the 194 multi-visit FAIB TW trees (292 transitions, mean interval 13.1 yr); the FAIB-fitted mean annual survival (0.994/yr) is slightly lower than the Busing-calibrated value (0.998/yr) and does not alter the qualitative conclusions. FAIB-fitted growth-class advancement rates are slower than the Busing model, likely reflecting that the FAIB sample spans degraded as well as prime old-growth stands; Busing-calibrated growth rates are retained for projections (full comparison in [results/analysis/faib_vital_rates.json](#)). The baseline annual burn probability used in fire scenarios was updated from a placeholder (0.002/yr) to the BC fire-history estimate of 0.000794/yr (96,543 ha burned across the 980,000 ha study area over 1900–2024); this lower rate reinforces, rather than changes, the conclusion that historical fire rates alone are insufficient to drive population-level decline without concurrent logging. In intact, protected old-growth the modelled stand is viable ($\lambda = 1.02$, quasi-extinction probability ≈ 0 over 200 yr), consistent with Busing & Spies' conclusion that undisturbed stands are stable to slowly growing. Critically, the matrix elasticities are highest for recruitment and growth through the small and mid-size classes (stages 4–5: 0.30, 0.28) and lowest for large-tree survival (stages 6–7: 0.06, 0.02), identifying recruitment, not adult survival, as the demographic lever for stand-level viability. The two best-documented stressors act on exactly this lever. Ungulate browse — which removes 60–80% of seedlings and saplings in high-density areas (§2.6.4) — drops the modelled growth rate below replacement on its own ($\lambda = 0.99$, quasi-extinction probability 12%); combined with continued attrition of the large, bark-rich seed-producers (whose reproductive value, 8.5 and 8.2, is the highest of any stage), it drives quasi-extinction probability to 42% over 200 yr. By contrast, large-tree removal *alone* barely changes λ (1.02), because yew's demography is bottom-weighted — but those large trees are the dominant seed source (fecundity scales with stem diameter; Busing & Spies 1995), so their depletion erodes the very recruitment the population depends on. The large-tree deficit in the size structure is therefore both a record of past harvest and a brake on future recovery. This stand-scale viability result complements the landscape projection (Figure 11): logging removes the old-growth canopy that yew recruitment requires, while browse suppresses recruitment where canopy persists — both converging on the same recruitment bottleneck, and both pointing to maintenance of old-growth shade and ungulate management as the decisive conservation levers (§4.3).

4. Discussion

4.1 A decline that likely understates true losses

The 67.9% overall decline reported here is a lower-bound estimate conditional on the sampled tiles, not a validated province-wide figure. Four sources of conservatism are worth noting. First, the analysis does not model future logging under existing tenure commitments, which will continue to reduce old-growth area. The true province-wide decline across the full ~3.8 million hectares of CWH+ICH+CDF zone would require systematic coverage but would almost certainly not be lower than the 67.9% reported here. Second, the logging-loss estimate does not incorporate indirect recruitment suppression by ungulates and bud mites. Both Vancouver Island and Haida Gwaii host overabundant deer populations following the extirpation of wolves and cougars from many coastal islands (TODO cite), and ungulate browse pressure on yew seedlings and saplings (0–10 cm DBH) has been documented to cause 60–80% seedling mortality in high-density areas, creating a regeneration debt that does not appear in canopy-cover mapping. The comparison between the field sample (fewer small-diameter stems, median 12.7 cm) and the FAIB inventory (more abundant small stems, median 5.5 cm; §3.3) is consistent with ungulate pressure or *Cecidophyopsis psilaspis* mite suppression being spatially heterogeneous — more severe in the coastal field sites than in the broader FAIB plot network. Third, our fire accounting (~700 ha in the sequential pipeline) underestimates fire's historical footprint: applied to the full original habitat independent of logging, fire perimeters would have suppressed an estimated ~3,730 ha (2.4% of original), ~3,330 ha of it in areas also affected by logging (§3.2.2). Fourth, the study tiles were not randomly selected and may undersample the most heavily logged or most remote parts of the province.

4.2 Two independent lines of evidence converge

The remotely sensed habitat decline, the field-measured population size structure, and the provincial FAIB PSP inventory are methodologically independent — derived from different data sources, collection protocols, and spatial scales — yet all three converge on the same diagnosis: a population that has lost a large fraction of its historical extent and whose remaining individuals show a structural deficit in the size classes responsible for reproduction and bark-yield. The depleted large-tree cohort (>30 cm DBH; §3.3) corresponds to trees of 150–700+ years of age — exactly the cohort eliminated by commercial logging of old-growth stands and targeted taxol bark extraction. A viable self-sustaining population would show a reverse-J size distribution; instead, the field sample gives $q \approx 2.0$ (95% CI 1.6–2.7) and the larger FAIB dataset gives $q \approx 2.7$ (95% CI 2.1–3.7), both substantially steeper than the stable-population reference $q = 1.5$. This three-way convergence strengthens the evidentiary basis for the conservation concern raised by the remote sensing results alone, and underpins the IUCN assessment in §4.6.

Looking forward, the consequences of these size-structure deficits can be projected with the Busing & Spies (1995) stage-projection matrix for old-growth yew (dominant eigenvalue $\lambda = 1.02$; Figure 11). Because recruitment into the large-diameter classes is intrinsically slow, a stand stripped of its >5 cm stems rebuilds that cohort within ~40 years but does not restore the >25 cm bark- and seed-producing trees for ~140 years — consistent with the field observation that such trees require centuries to attain harvest size (Graham 1994). The reproductive implication is central: because yew fecundity scales with stem diameter (Busing & Spies 1995), the depleted large-tree cohort represents the loss of the population's dominant seed producers, not merely its oldest individuals, so the structural deficit is also a recruitment deficit. At the province scale the trajectory is therefore decision-dependent (Figure 11): protecting the remaining habitat and allowing size structure to recover could sustain of the order of one to two million mature individuals over the coming two centuries, whereas continued logging of the 89% of mapped habitat that lies outside protected areas (§4.3) would reduce the mature population by roughly an order of magnitude.

4.3 Protected areas offer insufficient refuge

Only 5.6% of mapped yew habitat falls inside provincial parks (2,121 ha of 37,858 ha at $P \geq 0.5$), rising to 11.0% (4,180 ha) when conservancies (1,602 ha, 4.2%) and national parks (457 ha, 1.2%) are included (Figure S3). Nearly half of all mapped habitat (17,785 ha, 47%) lies in tiles with no protected area at all, including the largest individual yew concentrations: Alberni Valley (3,632 ha), Port Hardy Forest (1,173 ha), Blunden Harbour (1,085 ha), and Jervis Inlet Slopes (984 ha); the geographic concentration of this unprotected habitat on the outer and central coast is shown in Figure 7. This protected fraction falls far short of the 30% target of the Kunming-Montreal Global Biodiversity Framework (2022), and is particularly concerning given that the largest unprotected concentrations lie in active timber supply areas with no current regulatory constraint on old-growth harvesting.

Priority conservation actions include: (1) incorporation of Pacific yew habitat into forest stewardship plans for CWH and ICH operating areas; (2) riparian buffer widening to protect streamside populations from logging-driven erosion; (3) ungulate management in high-value yew stands to permit natural regeneration; (4) monitoring for *Cecidophyopsis psilaspis* spread in coastal populations; and (5) long-term fire management planning that recognises yew's zero fire tolerance.

4.4 Compounding threats amplify logging impacts

The threats to Pacific yew are not independent, logging creates the conditions for other threats to intensify. Canopy removal increases peak flows, eroding the riparian zones where yew is most abundant (§3.2.3; Hartman & Scrivener 1990). Clearcuts attract ungulates to forest edges, increasing browse pressure on any yew recruits attempting to establish. Forest fragmentation creates drier microclimates that raise fire risk. Stressed, isolated populations may also show reduced resistance to *Cecidophyopsis psilaspis*. The result is that the logging loss estimates in §3.2 understate the full cascade of harm: even old-growth patches that remain unlogged lose ecological value when surrounded by clearcut matrix that promotes edge effects, elevates deer density, and reduces moisture retention.

4.5 Limitations

1. Spectral similarity: The classifier cannot distinguish Pacific yew understorey from spectrally similar moist-forest conditions; probabilities represent habitat suitability rather than confirmed presence. A probability surface was used in preference to a binary map because it is threshold-free and propagates uncertainty throughout the analysis (§2.5.4).
2. VRI accuracy: Stand-age assignments have known errors, particularly for multi-cohort stands and post-fire regeneration, which could lead to misclassification of some second-growth as old-growth or vice versa.
3. Spatial coverage: the 98 tiles are a purposive (non-random) sample selected to span the diversity of BC yew habitat (§2.1), not a probability sample of the provincial range; province-wide extrapolation therefore requires caution and the headline decline figure is reported as conditional on the sampled tiles (§4.1).
4. Static analysis: The study represents a snapshot as of 2024; ongoing logging and climate change will alter the estimates.
5. No systematic field validation: Model predictions have not been validated against independent field surveys across the full study extent. The crowd-sourced field reporting interface on the web map (§2.8) could enable community-based validation over time.

A natural extension that would integrate the two halves of this study — the remote-sensing habitat surface and the stand-level demography (§3.3, Figure S5) — is a spatially explicit integral projection model (IPM) that

couples local vital rates to the mapped habitat. IPMs parameterised from repeat airborne LiDAR have recently been used to estimate size-dependent growth, survival, and recruitment of forest trees at landscape scale (e.g. remote-sensing demography from multi-temporal LiDAR; TODO cite), providing a template for propagating the viability analysis here across the full mapped extent and projecting yew abundance under spatially varying logging, browse, and fire pressure rather than at a single representative stand.

4.6 IUCN Red List Assessment for the Canadian (British Columbia) Population

The current global IUCN Red List status of *Taxus brevifolia* is Near Threatened (NT; Thomas 2013), an assessment that pre-dates any spatially explicit, province-scale analysis of BC habitat extent or decline. Using the quantitative results of the present study, we apply IUCN Red List criteria (IUCN 2012) to the Canadian BC population, which represents the northern core of the species' global range and the largest temperate rainforest population.

Criterion A — Population size reduction inferred from habitat decline

Criterion A2c evaluates population size reductions estimated or inferred where the reduction or its causes may not have ceased, based on observed decline in habitat area, extent, or quality. Pacific yew reaches reproductive maturity at approximately 80–100 years (Graham 1994); three generations therefore span ~240–300 years, well encompassing the industrial logging era beginning in the 1920s. Our analysis documents a 67.9% observed habitat decline across 9,800 km² of sampled BC range over approximately the past 100 years (the period of intensive industrial logging from the 1920s through the present). This is a partial-window observation, not a 3-generation projection: the IUCN A2c criterion permits using a shorter observed window when the causes of decline are ongoing and inferred to have persisted for at least 1 generation. Because industrial logging began in the 1920s and continues under existing tenure commitments, the 3-generation extrapolation would equal or exceed the observed 67.9% figure. A full 3-generation projection using the Busing & Spies stage-projection matrix seeded with current habitat extent and logging rates would require province-wide coverage beyond the 98 studied tiles; we report the 67.9% as a conservative lower bound on 3-generation decline rather than a final estimate.

IUCN Criterion	Threshold	This Study	Assessment
CR A2c	≥ 80% reduction	99.1% (CDF); 74.7% (ICH)	CR at subpopulation level
EN A2c	≥ 50% reduction	67.9% (BC overall)	EN at population level
VU A2c	≥ 30% reduction	Exceeded	—

A 67.9% estimated habitat decline where the cause is ongoing and not reversible is consistent with the Endangered (EN) threshold (≥50%), recognising that this inference rests on modelled habitat rather than a direct population census. The 99.1% CDF decline and 74.7% ICH decline each individually exceed the Critically Endangered threshold (≥80%) at the subpopulation level.

Criterion A2d — Levels of exploitation

The taxol bark harvest of 1989–1993 constituted direct exploitation at a scale unparalleled in the species' recorded history, with an estimated 360,000 mature trees per year felled across the Pacific Northwest at peak harvest (Hartzell 1990). Although bark harvest has ceased, its legacy is embedded in the depleted large-tree cohort documented in §3.3, providing independent field-based corroboration of the A2c finding. This reinforces the EN designation under criterion A2d.

Criterion B — Geographic range

Criterion B1 (EOO) does not apply; the EOO substantially exceeds the 5,000 km² EN threshold. The remaining modelled yew habitat totals ~497 km² across the 98 tiles, approaching the EN B2 AOO threshold of <500 km². However, the tiles cover only ~0.26% of the CWH+ICH+CDF zone extent; the 98-tile AOO cannot be extrapolated to a true provincial AOO without systematic province-wide coverage, and Pacific yew does not formally qualify for increased protection under this criterion.

Criteria C, D, and E

Analysis of the 461 deduplicated yew stems from 122 FAIB PSP plots gives a conditional mature-stem density (DBH ≥ 10 cm) of 10 stems/ha (median; IQR 4.9–24.7 stems/ha) in plots where *T. brevifolia* was recorded. Because the 49,663 ha probability-mass habitat estimate is a probability-weighted area ($\Sigma p \times 0.01$ ha/pixel; §2.5.4) it already accounts for partial occupancy, so multiplying by the conditional density gives an order-of-magnitude population estimate of approximately 497,000 mature individuals (range 243,000–1,227,000 across the IQR). This substantially exceeds the <2,500 EN threshold under criterion C. For context, the Busing & Spies (1995) stage-projection model of undisturbed old-growth stands in Oregon and Washington gives much higher densities — ~300–500 stems/ha total and ~40–60 stems/ha >5 cm DBH — so the FAIB-calibrated estimate used here is conservative: FAIB plots sample the full range of BC stand conditions (including marginal stands), whereas the Busing values describe prime occupied old-growth. Applying the Busing old-growth density to the remaining habitat would raise the estimate severalfold; we retain the FAIB-based figure as a lower bound. Criteria D and E are not met.

Summary assessment

Under the best available evidence, the Canadian BC population of *Taxus brevifolia* supports a classification consistent with Endangered (EN A2c) under IUCN Red List criteria, recognising that this inference rests on modelled habitat decline rather than direct population census. This conclusion is independently corroborated by the field-measured population size structure (§3.3; §4.2). The CDF and ICH subpopulations individually reach Critically Endangered thresholds. The current global designation of Near Threatened (Thomas 2013) is not consistent with the magnitude of BC habitat decline documented here, nor with the ongoing nature of the primary threat. We recommend that COSEWIC and the IUCN undertake a formal reassessment of *Taxus brevifolia* using province-scale spatial data.

5. Conclusions

This study presents the first spatially explicit, province-scale assessment of Pacific yew (*Taxus brevifolia*) habitat extent and decline in British Columbia, using 64-dimensional satellite spectral embeddings from the Google AlphaEarth Foundation model classified by an XGBoost ensemble (AUC-ROC 0.996). Across 9,800 km² of sampled CWH, ICH, and CDF forest, we estimate a 67.9% decline from approximately 154,483 ha of original habitat to 49,663 ha remaining — driven overwhelmingly by industrial clear-cut logging, with secondary contributions from wildfire, riparian erosion, and the legacy of taxol bark extraction. Calibrated against 461 deduplicated FAIB permanent sample plot records (conditional mature-stem density: median 10 stems/ha, IQR 4.9–24.7), the remaining 49,663 ha supports an estimated ~497,000 mature individuals. Both the field-measured sample ($n = 120$; binomial $p \approx 4 \times 10^{-9}$) and the FAIB inventory ($n = 461$ deduplicated stems; $p \approx 1.2 \times 10^{-43}$) independently confirm severe depletion of the large-adult cohort (DBH > 30 cm) relative to the de Liocourt stable-population model, consistent with selective harvest of large individuals during the 1989–1993 taxol rush.

The finding that only 11.0% of mapped yew habitat falls within protected-area designations underscores the species' extreme exposure to continued forestry operations. Application of IUCN Red List criteria to the

Canadian BC population supports a classification consistent with Endangered (EN A2c), a significant upgrading from the current global Near Threatened listing, with the CDF and ICH subpopulations individually meeting Critically Endangered thresholds. We urge a formal COSEWIC and IUCN reassessment of the species drawing on provincial-scale spatial data, and recommend that Pacific yew habitat be explicitly incorporated into forest stewardship plans across the Coastal Western Hemlock and Interior Cedar–Hemlock zones.

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Data Availability

The interactive web map is available at jerichooconnell.github.io/yew_project. Source code, model weights, and analysis scripts are hosted on GitHub. Satellite embeddings were accessed via Google Earth Engine; the VRI 2024 dataset was obtained from the BC Data Catalogue.

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Figure Captions

Main-text figures (11):

Figure 1. Estimated historical versus current remaining Pacific yew habitat by major BEC zone (dumbbell plot, log x-axis): the faded marker is estimated original habitat, the solid marker current remaining habitat, and the annotation the percentage decline. CWH retains the most in absolute terms; CDF has declined ~99%.

Figure 2. Post-classification suppression waterfall: raw model prediction reduced through logging, fire, and elevation filters to the final current-habitat estimate across all 98 study tiles.

Figure 3. Location map of the 98 study tiles across British Columbia (coastal CWH/CDF tiles = circles; ICH interior tiles = triangles), overlaid on biogeoclimatic zone boundaries.

Figure 4. Classifier performance comparison for six models trained on the 64-dimensional AlphaEarth satellite embeddings (held-out validation set). AUC-ROC is shown alongside accuracy; the production XGBoost model (AUC 0.996) is statistically indistinguishable from the best neural models.

Figure 5. Single-tile threat diagnostics for one representative CWH tile (Sechelt Peninsula, $\sim 10 \times 10$ km): (a) RGB context; (b) raw yew habitat probability; (c) VRI stand-age (logging) classes; (d) historical fire perimeters by recency; (e) 30 m riparian-erosion buffer intersected with old-growth habitat; (f) protected areas over mapped habitat. Predicted habitat (b) tracks the riparian corridor, and only $\sim 1\%$ of the tile's habitat falls inside a protected area.

Figure 6. Predicted Pacific yew probability shown as the actual per-tile model-output rasters placed at their true geographic locations on Vancouver Island (inset: location within BC). Each labelled patch is one $\sim 10 \times 10$ km study tile, coloured by suppressed yew probability; high-probability habitat persists in old-growth-rich tiles (e.g. Carmanah-Walbran, Port Renfrew) while heavily logged tiles show sparse, fragmented predictions.

Figure 7. Geographic distribution of mapped Pacific yew habitat across the 98 study tiles, each symbol sized in proportion to its mapped habitat area ($P \geq 0.5$) and coloured by the percentage of that habitat inside protected areas (exposed $\rightarrow$ protected). Most high-habitat tiles, particularly on the outer and central coast, are largely unprotected.

Figure 8. Geography of four key tile-level variables across the 98 study tiles on a common base map, marker area proportional to mapped habitat: (a) mapped habitat ($P \geq 0.5$); (b) percentage of forest area logged; (c) percentage old-growth remaining; (d) percentage of habitat inside protected areas. Logging concentrates in the southern coast and interior; old-growth refugia persist on the central and north coast; protection is sparse and spatially mismatched with the largest habitat concentrations.

Figure 9. Pacific yew population size-class distribution (DBH converted from field-measured CBH, $n = 120$). Left: observed stem counts per 10-cm DBH class versus the de Liocourt stable-population reference ($q = 1.5$) and the reverse-J slope fitted to the whole observed distribution ($q \approx 2.0$, 95% CI 1.6–2.7). Shaded area: the large-tree deficit (>30 cm DBH, 7 observed vs. ~ 32 expected). Right: observed versus expected cumulative distribution functions, with the observed median (12.7 cm) marked.

Figure 10. De Liocourt size-structure comparison: FAIB PSP inventory ($n = 431$ live stems, deduplicated) versus the field sample ($n = 120$). Left: FAIB observed counts per 10-cm DBH class versus the $q = 1.5$ reference and the whole-population fitted slope ($q = 2.73$, 95% CI 2.13–3.70). Centre: equivalent analysis for the field sample ($q = 2.00$, CI 1.63–2.69). Right: distribution of conditional TW stem density (PHF_TREE expansion) across FAIB plots with *T. brevifolia* present; the median 10.0 mature stems/ha (DBH ≥ 10 cm) underpins the population estimate (§4.6). Both datasets show significant large-tree depletion, the FAIB signal stronger owing to sample size.

Figure 11. Predicted Pacific yew trajectories under best- and worst-case scenarios. (a) Stand-level post-harvest recovery from the Busing & Spies (1995) stage-projection matrix ($\lambda = 1.02$): all harvestable stems (>5 cm DBH) recover to 90% of old-growth density in ~ 40 yr, but the large-tree cohort (>24.9 cm DBH — the bark/seed producers) takes ~ 140 yr. (b) Province-scale mature yew (DBH ≥ 10 cm), 2024–2224: best case (remaining habitat protected, size structure recovers) versus worst case (the 89% of habitat currently outside

protected areas logged by 2124) and a status-quo trajectory; bands span the FAIB-to-Busing density range. Illustrative scenarios, not a calibrated forecast.

Supplementary figures (4):

Figure S1. Stacked bar chart of destroyed versus remaining yew habitat in each CWH subzone, ordered by total estimated original habitat.

Figure S2. Stacked bar chart of destroyed versus remaining yew habitat in each ICH subzone.

Figure S3. Where the mapped yew habitat sits by protection status: provincial parks (5.6%), conservancies (4.2%), and national parks (1.2%) together protect only 11.0%, leaving 89% unprotected; a large share lies in tiles with no protected area at all.

Figure S4. As Figure 6, for the Central and North Coast — the region holding the largest remaining yew habitat. Substantial predicted habitat persists in less-accessible outer-coast and fjord tiles (e.g. Rivers Inlet, Smith Sound), consistent with lower historical logging pressure.

Figure S5. Stand-level population viability analysis from the Busing & Spies (1995) stage-projection matrix seeded with the observed yew size structure (FAIB + field). (a) Projected stand population relative to today (log scale) under four scenarios: protected old-growth ($\lambda = 1.02$), ungulate browse removing 70% of recruitment ($\lambda = 0.99$), continued large-tree harvest at 2%/yr ($\lambda = 1.02$), and browse + harvest combined ($\lambda = 0.99$). (b) Quasi-extinction probability (population falling below 10% of current within 200 yr; 500 stochastic replicates, vital-rate CV = 0.15): negligible under protection, 12% under browse alone, and 42% under browse + harvest. Matrix elasticities are highest in the small/mid size classes, identifying recruitment — not adult survival — as the demographic lever.